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N[6]-methyladenine modification of DNA enhances RecA-mediated homologous recombination

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Both DNA methylation and homologous recombination (HR) are extensively studied. In bacteria, Dam methylation is the most studied DNA modification, while RecA-mediated HR is a primary mechanism to repair DNA damages including double-stranded breaks, single-stranded gaps, and stalled replication forks. While HR regulation by proteins is extensively studied, whether methylation of DNA itself directly affects the functions of RecA and HR remains unclear. Mainly by single-molecule experiments, we report that Dam methylation of single-stranded DNA (ssDNA) promotes RecA assembly, partially by reducing the effective charge of ssDNA under counterion screening. Furthermore, Dam methylation of double-stranded DNA promotes homologous pairing, joint molecule growth, and strand exchange. In cellular experiments, *dam* deletion impairs HR, whereas hypermethylation of the adenines in the genome enhances HR in P1 transduction assays and DNA-damage sensitivity tests without significantly upregulated HR-related genes. In addition, the preference of RecA for Dam-methylated DNA in RecA assembly and homologous pairing is conserved across divergent species covering a gram-negative bacterium *Klebsiella pneumoniae*, a gram-positive bacterium *Bacillus subtilis,* and a flowering plant *Arabidopsis thaliana*. Dam methylation of ssDNA increases the ATPase activity of molecular motors such as RecQ helicase that containing RecA-like domains. These findings reveal effects of DNA methylation and mechanisms regulating RecA-mediated HR and molecular motors.

DNA methylation | homologous recombination | single-molecule | RecA | bacteria

Significance

DNA methylation is well known for its roles in gene regulation and mismatch repair. However, its influence may extend to other fundamental DNA metabolic processes. This study uncovers the role of DNA methylation in homologous recombination, a critical mechanism for maintaining genomic stability. By elucidating how DNA methylation modulates the activity of RecA proteins across diverse organisms, we reveal an expanded connection between epigenetic regulation and recombination-based DNA repair. These findings not only advance our understanding of genome maintenance but also offer unique perspectives for combating bacterial resistance to genotoxic stress.

DNA methylation is a key epigenetic mechanism regulating fundamental cellular processes across species (1, 2). DNA adenine methyltransferase (Dam) is the most studied bacterial DNA-methyltransferase, adding methyl groups to adenines (m6A) at GATC sites (3). Dam methylation participates in processes including antiviral defense, cell cycle regulation, DNA replication, transcriptional regulation, and methyl-dependent mismatch repair (MMR) (2–9). Homologous recombination (HR) is essential for the *dam*-knockout *Escherichia coli* (*Ec*), as indicated by poor viability of double-knockout strains lacking *dam* and an HR-related gene such as *recA*, *recB*, *recC*, *uvrA,* or *recG* (7, 10–13).

Double-strand breaks (DSBs) are the most detrimental DNA damage, leading to genome instability, mutations, and cell death. Since bacteria lack nonhomologous end-joining, they rely on HR to repair DSBs with high fidelity by using a homologous double-stranded DNA (dsDNA) template (14). Besides DSBs, HR also plays essential roles in repairing single-stranded gaps and stalled replication forks. The bacterial RecA, archaeal RadA, and eukaryotic Rad51 recombinases play key roles in HR. In *Ec*, RecBCD initiates DSBs repair by binding to DNA ends, unwinding, and asymmetrically degrading both strands. Upon encountering a Chi site, RecBCD generates single-stranded DNA (ssDNA) with 3′overhangs (15). The ssDNA-binding protein (SSB) binds rapidly to ssDNA, preventing DNA secondary structures that could inhibit RecA assembly. RecA has two DNA-binding sites (16, 17). The first site is involved in displacing SSB from the ssDNA and taking its place (18). RecOR can assist RecA assembly on SSB-coated ssDNA (19–23).

After forming RecA-ssDNA filaments, RecA uses its second, lower-affinity site to bind homologous dsDNA during homologous pairing (14, 15). The filament aligns the incoming strand with the complementary strand of dsDNA, displacing the outgoing strand to create a displacement loop during the strand exchange process (16, 24, 25), followed by DNA synthesis and recombination resolution (24–28). RecA requires ATP but not ATP hydrolysis to assemble on ssDNA and mediate homologous pairing (16, 24–28). ATP hydrolysis is required to bypass sequence heterology during strand invasion (29) and

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promote dissociation of RecA from DNA once strand invasion is completed (25, 30), which is essential for successful HR. Although the regulation of RecA by cofactors such as proteins is well studied (16, 31, 32), whether methylation of DNA itself directly regulates the functions of RecA remains unclear.

Single-molecule experiments have provided dynamic insights into RecA assembly, complementing biochemical and genetic studies (28). Single-molecule fluorescence resonance energy transfer (smFRET) has been used to study RecA assembly on short ssDNA (33), SSB displacement (34), and the coupling of ATP hydrolysis with filament dynamics (35). Single-molecule fluorescence microscopy has been used to observe the slow nucleation and subsequent fast growth of RecA on SSB-coated ssDNA, facilitated by RecFOR (23). Magnetic tweezers (MT) have been used to reveal the force and ATP hydrolysis dependence of RecA assembly (36).

Homologous pairing and subsequent steps have been explored using single-molecule experiments as well (37, 38), including studies of salt dependence (39), and mismatch sensing during strand exchange (40) using MT and smFRET. The smFRET has been used to study RecA-ssDNA filament sliding on dsDNA (41) and the lifetime of joint molecules during homologous search (42). MT and smFRET have been used to observe strand exchange in real-time (43, 44). Tethered particle motion (TPM) has been used to reveal the 5′-to-3′ polarity of strand exchange (45). These single-molecule studies have illuminated the functions and regulation of RecA, providing insights into HR.

Results

Methylation of ssDNA Promotes RecA Assembly. In cells, HR cannot begin until RecA assembly on ssDNA, so we first studied this process. In MT experiments, we tethered dsDNA between a paramagnetic bead and the glass surface. Following a previous study (46), we generated stretched ssDNA by holding the molecules at about 40 pN in the presence of 100 mM NaOH, which allows the untethered strand in dsDNA to be peeled off (*SI Appendix*, Fig. S1). Although DNA overstretching typically occurs above about 65 pN in physiological buffers, the alkaline condition destabilizes the double helix and enables strand separation at lower forces. Thereafter, we exchanged the buffer for RecA reaction buffer to proceed with the RecA assembly assays. The ssDNA can form secondary structures under forces less than 8 pN (36, 47), and the binding of SSB to ssDNA can prevent these DNA secondary structures (30, 34, 48, 49). To avoid secondary structures, we studied the assembly of RecA on naked 6 k-nt ssDNA at 8 pN and on SSB-coated ssDNA at 3 pN. The 6 k-nt ssDNA contains 22 GATC sites, which are recognized by Dam methyltransferase for methylation, and a total of 1,347 adenines available for m6A modification. As the extension of RecA-ssDNA filament is longer than that of SSB-coated and naked ssDNA (Fig. 1*A*), we monitored the assembly of RecA by observing the elongation of ssDNA in real-time, similar to the previous MT works (30, 36, 50). In Fig. 1*A*, the concentrations of 2 μM RecA and 200 nM SSB are higher than those required for saturated binding (*SI Appendix*, Fig. S2). We studied the RecA assembly on SSB-coated ssDNA (200 nM SSB) with ATP (Fig. 1*B*) or ATPγS (Fig. 1*C*), with ATP in the presence of RecOR (Fig. 1*D*), and on naked ssDNA with ATP (Fig. 1*E*). ATPγS is an ATP analog with negligible hydrolysis activity. When bound to RecA, ATPγS stabilizes the filament by preventing ATP hydrolysis-dependent disassembly (36, 51). For each condition, we obtained average curves, with error bars shown as shadows, from more than three DNA molecules (*SI Appendix*, Fig. S3).

Fig. 1*B* shows that RecA replaces SSB faster from Dam-m ssDNA and even faster from m6A-m ssDNA than from non-m ssDNA. At 300 nM, RecA can replace SSB from m6A-m ssDNA but not from Dam-m or non-m ssDNA. We calculated the initial rate of RecA assembly at 600 nM with a previously used method (48). The initial rate for RecA assembly ($0.62\ \mathrm{nm \cdot s^{-1}}$) is similar to previous work (23), and the initial rate increased by about 1.8-fold for Dam-m ssDNA and about 7.1-fold for m6A-m ssDNA, respectively. At 1,200 nM, DNA methylation also speeds up RecA assembly, and non-m, Dam-m, and m6A-m ssDNA achieve the same DNA extension after reaction. Fig. 1*C* shows that, with ATPγS, m6A and Dam methylation speed up the replacement of SSB, similar to the situation with ATP. At 600 nM RecA, the initial rate increased 1.9-fold for Dam-m ssDNA and 5.9-fold for m6A-m ssDNA. RecO and RecR can facilitate RecA assembly on SSB-coated ssDNA (19–23). Following the concentration ratio of the previous work (23), we applied a 10:1:1 ratio for RecA, RecO, and RecR in Fig. 1*D*, where 80, 150, and 300 nM RecA were used, respectively. We found that, with RecO and RecR, about 1/4 concentration of RecA is required for SSB displacement compared to conditions without them, where 300, 600, and 1,200 nM RecA were used. At 150 nM RecA, the initial rate increased by about 1.8-fold for Dam-m ssDNA and about 7.6-fold for m6A-m ssDNA, respectively (Fig. 1*D*).

Fig. 1*E* shows that DNA methylation also promotes RecA assembly on naked ssDNA. RecA/Rad51 assembly involves a slow nucleation phase followed by rapid filament growth (16, 23, 30, 52). Using Monte Carlo calculations, we fitted the assembly curve at 200 nM RecA (*SI Appendix*, Fig. S4), as done previously for RecA and Rad51 (30, 46, 53). We found Dam methylation speeds up nucleation by 2.0-fold and growth by 1.9-fold, while m6A methylation further speeds up nucleation by 5.1-fold and growth by 3.2-fold. The nucleation ($1.2 \times 10^{-3}\ \mathrm{nt^{-1}\ s^{-1}}$) and growth rates ($0.35\ \mathrm{s^{-1}}$) on non-m ssDNA are in agreement with previous results (30).

To assess the impact of Dam methylation of a single GATC site on RecA nucleation, we used a 20-nt ssDNA containing only one GATC site in biolayer interferometry (BLI) experiments (Fig. 1*F*), similar to previous Rad51 studies (54). RecA nucleation on even shorter ssDNA was difficult to analyze quantitatively (*SI Appendix*, Fig. S5). Global fitting (black dashed lines) showed Dam methylation accelerates the nucleation of RecA on ssDNA (k_{on}) by about 1.8-fold whereas barely affecting its dissociation from ssDNA (k_{off}). Methylating all five adenines in the same sequence led to a greater increase of about 3.9-fold in k_{on} (*SI Appendix*, Fig. S6).

In BLI experiments, we used gradient repetition with three RecA concentrations. Although data from a single RecA concentration are sufficient to calculate one set of k_{on} and k_{off}, a shared parameter set fits the experimental curves well across all three concentrations (Fig. 1*F*). BLI instruments are optimized for simple two-state binding and dissociation kinetics. However, in the presence of ATP hydrolysis, RecA filaments exhibit complex dynamics that deviate from this classical model. ATP-bound RecA promotes filament formation, while ATP hydrolysis to ADP destabilizes the filament, leading to subunit dissociation. These processes occur asynchronously and introduce nonequilibrium behavior (23, 55). As a result, under steady-state ATP conditions, the BLI signal reflects contributions from multiple dynamic states, complicating global fitting of binding and dissociation phases and preventing accurate extraction of the k_{on} and k_{off} values. When we used ATP, despite observing faster assembly of RecA on Dam-methylated ssDNA, BLI failed to yield the k_{on} and k_{off} values

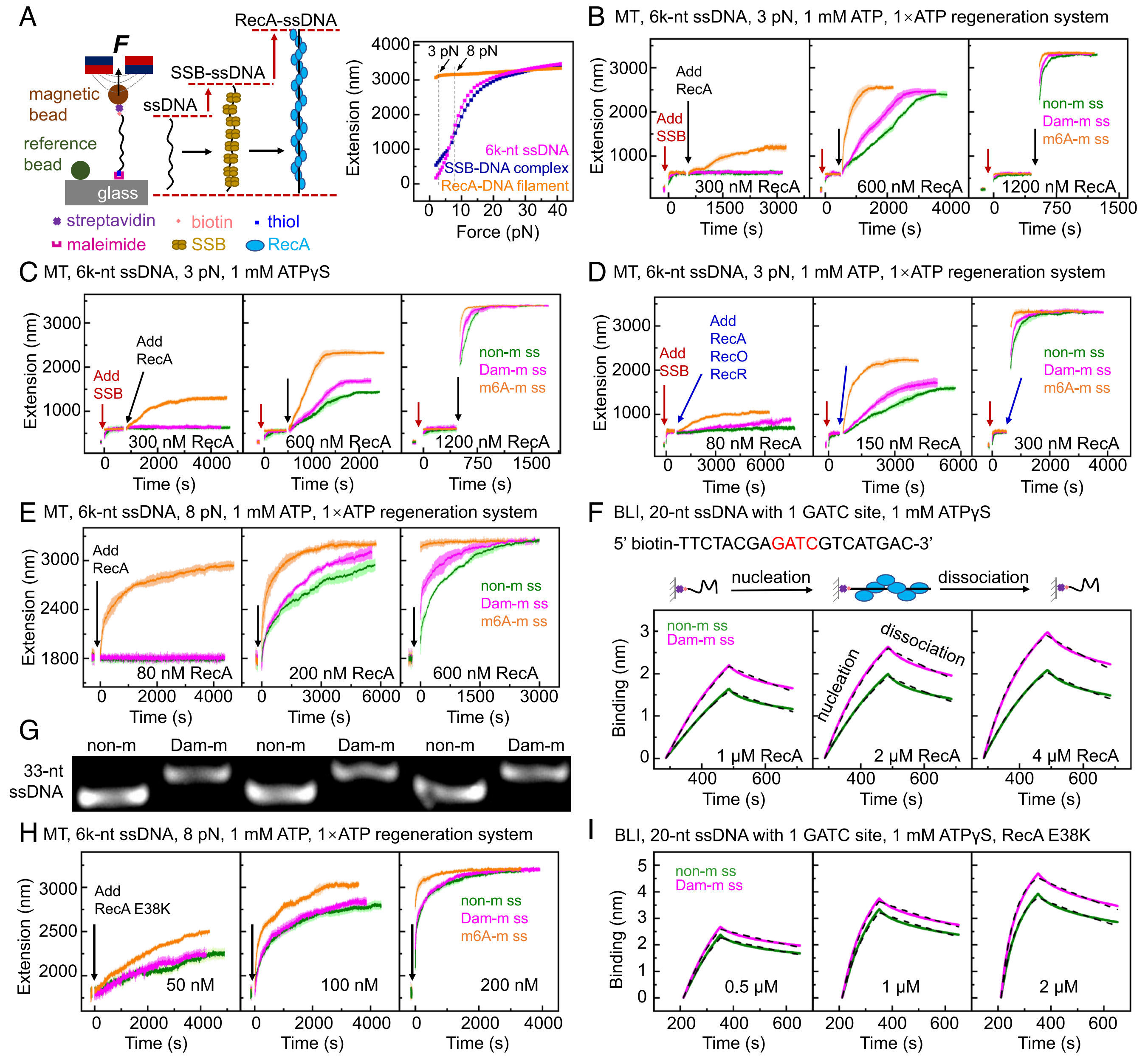

Fig. 1. RecA assembles faster on methylated ssDNA. In MT experiments, we used a 6 k-nt ssDNA containing 22 GATC sites. In BLI experiments, we used a 20-nt ssDNA containing only one GATC site. (*A*) In MT experiments, RecA assembly leads to ssDNA elongation. (*B* and *C*) RecA assembly on SSB-coated ssDNA with ATP or ATPγS. (*D*) RecA assembly on SSB-coated ssDNA with ATP in the presence of RecO and RecR. (*E*) RecA assembly on naked ssDNA with ATP. (*F*) RecA nucleation with ATPγS in BLI experiments. (*G*) Dam methylation reduces electrophoretic mobility of ssDNA. See the original, uncropped gel images in the *SI Appendix*, Fig. S8. (*H*) Assembly of RecA E38K on ssDNA in MT experiments. (*I*) RecA E38K nucleation with ATPγS in BLI experiments. In panels (*B* to *E*, and *H*), shadows represent SE from more than three DNA molecules, and arrows indicate the addition of RecA or other proteins in MT experiments. In panels (*F* and *I*), global fitting curves for three RecA concentrations are plotted as black dashed lines.

(*SI Appendix*, Fig. S7). Therefore, we used ATPγS in our BLI experiments to stabilize RecA filaments and simplify kinetic analysis. All assembly parameters are summarized in *SI Appendix*, Table S1.

As RecA assembly is highly salt dependent (56), we examined how Dam methylation affects the effective charge of ssDNA under counterion screening. We found Dam methylation reduced the electrophoretic mobility of 33-nt ssDNA with three GATC sites by 3.7% ± 0.2% (Fig. 1*G* and *SI Appendix*, Fig. S8). Although Dam methylation does not alter the formal charge of the DNA backbone, it decreases the effective charge of ssDNA under counterion screening, likely reflecting changes in the local electrostatic environment or ion association dynamics around the methylated DNA.

To investigate whether Dam methylation-induced reduction in effective charge of ssDNA affects RecA assembly, we measured the assembly of a point mutant RecA E38K using MT experiments (Fig. 1*H*). As shown in previous studies, the E38K mutation replaces a negatively charged glutamate with a positively charged lysine, reducing electrostatic repulsion between RecA and DNA and thereby enhancing binding affinity (57–59). In MT experiments, we found that both Dam methylation and m6A methylation of ssDNA have much weaker impacts on the assembly of RecA E38K than wild-type (WT) RecA (Fig. 1*H*). The Monte Carlo calculations show that Dam methylation increases $k_{\mathrm{nucleation}}$ and k_{growth} of RecA E38K both by about 1.2-fold. The m6A methylation increases $k_{\mathrm{nucleation}}$ and k_{growth} of RecA E38K by about 2.0-fold and 2.3-fold, respectively. To compare, for WT RecA, Dam methylation increases $k_{nucleation}$ and k_{growth} by 2.0-fold and 1.9-fold, respectively; the m6A methylation increases $k_{nucleation}$ and k_{growth} of RecA by about 5.1-fold and 3.2-fold, respectively (*SI Appendix*, Table S1). Consistent with MT results, BLI experiments showed Dam methylation increased the k_{on} of RecA E38K by 1.2-fold only, compared to 1.8-fold for WT RecA (Fig. 1 and *SI Appendix*, Table S1). The m6A methylation increased the k_{on} of RecA E38K by about 2.0-fold, compared to about 3.9-fold for WT RecA (*SI Appendix*, Fig. S6). The results shown in Fig. 1 *H* and *I* are consistent with a picture that Dam methylation decreases the effective charge of ssDNA under counterion screening and promotes RecA assembly. We acknowledge that RecA E38K has generally higher affinity for ssDNA than WT RecA, regardless of the methylation status of ssDNA. Given its higher binding affinity, we used lower concentrations of RecA E38K in both MT and BLI experiments.

Methylation of dsDNA Promotes Homologous Pairing, Joint Molecule Growth, and Strand Exchange. Since homologous pairing is a rate-limiting step in HR (14, 15), we next studied whether Dam methylation of DNA affects homologous pairing. We used smFRET to study homologous pairing with a 21-nt ssDNA containing two GATC sites, following a previously established assay (32, 41–43). As ATP hydrolysis is not required for homologous pairing (60), ATPγS was used to maintain the filament in its active form, prevent filament disassembly, and stabilize the homologous pairing products (37, 38). As reported, the product of homologous pairing with ATPγS is a joint molecule bound by RecA (60).

We found that Dam methylation of ssDNA had no detectable effect on homologous pairing, regardless of whether the dsDNA was methylated (*SI Appendix*, Fig. S9). We next used a competitive binding assay to study whether Dam methylation of dsDNA affects homologous pairing (Fig. 2 *A* and *B*). Because we labeled the Cy5 groups at different positions of the Dam-m dsDNA and non-m dsDNA, the FRET efficiency was higher for the joint molecules containing Dam-m dsDNA (about 0.78 in Fig. 2*A*, about 0.92 in Fig. 2*B*) than that containing non-m dsDNA (about 0.22 in Fig. 2*A*, about 0.40 in Fig. 2*B*). We found only Dam-m dsDNA was paired at low RecA-ssDNA filament concentrations (0.5 and 1 nM). At higher concentrations (≥5 nM), non-m dsDNA was paired after Dam-m dsDNA saturation. Overall, RecA filaments preferentially paired with Dam-m dsDNA, regardless of whether the ssDNA was methylated.

To quantify the acceleration of homologous pairing by Dam methylation of dsDNA, we recorded its kinetics in real-time, following previous studies (32, 38, 41, 42, 54). We either added 5 nM RecA-ssDNA filaments to bind the surface-anchored dsDNA (Fig. 2*C*) (42) or added 1 nM dsDNA templates to surface-anchored RecA-ssDNA filaments (Fig. 2*D*) (32, 41). To quantify the number of joint molecules (N) in a single camera view, we recorded three-second videos at each time point (t) and counted N in each video (*SI Appendix*, Fig. S10). We found that no homologous pairing events occurred on nonhomologous dsDNA substrates, while homologous pairing events occurred on the homologous dsDNA, as indicated by obvious increase of N (Fig. 2 *C* and *D*), confirming the specificity for homologous dsDNA in homologous pairing. We fitted the data using an equation $N(t)=N_0[1\text{-exp}(-t/\tau)]$. Here, N_0 is the saturated number of joint molecules and τ is the mean waiting time for formation of a joint molecule. We found Dam methylation of dsDNA speeds up RecA-mediated homologous pairing by about 4.9-fold in dsDNA anchored experiments (Fig. 2*C*), and about 3.5-fold in filaments anchored experiments (Fig. 2*D*).

In cells, homologous pairing often requires replicated sister DNA, which is often hemimethylated (61). Thus, we also measured the kinetics of homologous pairing with hemi-m dsDNA (Fig. 2 *C* and *D*). In the dsDNA-anchored configuration (Fig. 2*C*), hemimethylation at GATC sites on either the complementary strand (orange) or the outgoing strand (purple) significantly enhanced homologous pairing compared to non-m dsDNA. Specifically, methylation on the complementary strand and the outgoing strand increased pairing by about 1.8-fold. A similar trend was observed in the filament-anchored configuration (Fig. 2*D*), where pairing was enhanced by about 2.1-fold and about 1.9-fold for complementary and outgoing strand methylation, respectively. MT experiments showed that free RecA in solution did not assemble on dsDNA in the reaction buffer of pH 7.5 (*SI Appendix*, Fig. S11), ensuring no interference with the homologous pairing reaction.

Next, we compared the growth of joint molecules formed by non-m and Dam-m dsDNA using a 412-bp segment from the replication origin (*oriC*) region of *Ec* in MT experiments (Fig. 2*E*). After RecA assembly on the homologous 412-nt ssDNA, we added the filaments and recorded the extension of dsDNA. We found that Dam methylation of dsDNA increased the joint molecule growth rate by about 2.3-fold with ATPγS (0.12 to 0.28 $nm \cdot s^{-1}$).

We then used a bulk assay to study the effect of Dam methylation on strand exchange (Fig. 2*F*). We employed a 50-bp dsDNA derived from the *oriC* region and a 90-nt RecA-ssDNA filament containing the sequence of the dsDNA. The complementary strand of the 50-bp dsDNA was labeled with the Cy3. After incubation for 3, 10, 30, and 100 min with ATP (*SI Appendix*, Fig. S12), native PAGE was performed, and the strand exchange proportion was quantified by the relative decrease in dsDNA band intensity in the electrophoretic mobility shift assay (EMSA). After 30 min of incubation, Dam methylation of dsDNA increased the strand exchange efficiency by about twofold, from 10.0% ± 1.5% to 20.0% ± 0.4%. A time-course bulk strand exchange assay at 3 to 100 min further confirmed that Dam methylation of dsDNA promotes strand exchange. The strand exchange efficiency with ATPγS was much lower than that with ATP, consistent with previous results showing that ATP hydrolysis facilitates strand exchange (55). Dam methylation also increased the exchange rate in the presence of ATPγS (*SI Appendix*, Fig. S13). Similar to the results of homologous pairing in smFRET experiments, the methylation of ssDNA had no apparent effect on strand exchange in the bulk assay (*SI Appendix*, Fig. S14).

Methylation Enhances HR in Cells. To test whether DNA methylation promotes HR in cells, we performed a P1 transduction assay (62, 63) to introduce kanamycin resistance (Fig. 3). HR efficiency was measured by the number of surviving bacteria on kanamycin plates. We deleted the *dam* gene (Δ*dam*), rescued it with a plasmid-borne *dam* gene, referred to as Δ*dam* (*dam*), or introduced a plasmid-borne *EcoGII* gene, referred to as WT (*EcoGII*). EcoGII is a nonspecific m6A methyltransferase (64). As shown in Fig. 3, HR efficiency was significantly reduced in Δ*dam*, while hypermethylated WT (*EcoGII*) showed even higher HR efficiency than WT. These results indicate that Dam methylation enhances HR, and m6A methylation further boosts HR in cells, which is consistent with the above results about the effect of DNA methylation on RecA assembly, homologous pairing, joint molecule growth, and strand exchange.

By P1 transduction assay, we found that WT (*EcoGII*), which is hypermethylated, showed even higher HR efficiency than WT. One possible concern is that this increased efficiency results from an elevated level of DNA damage in WT (*EcoGII*), providing more

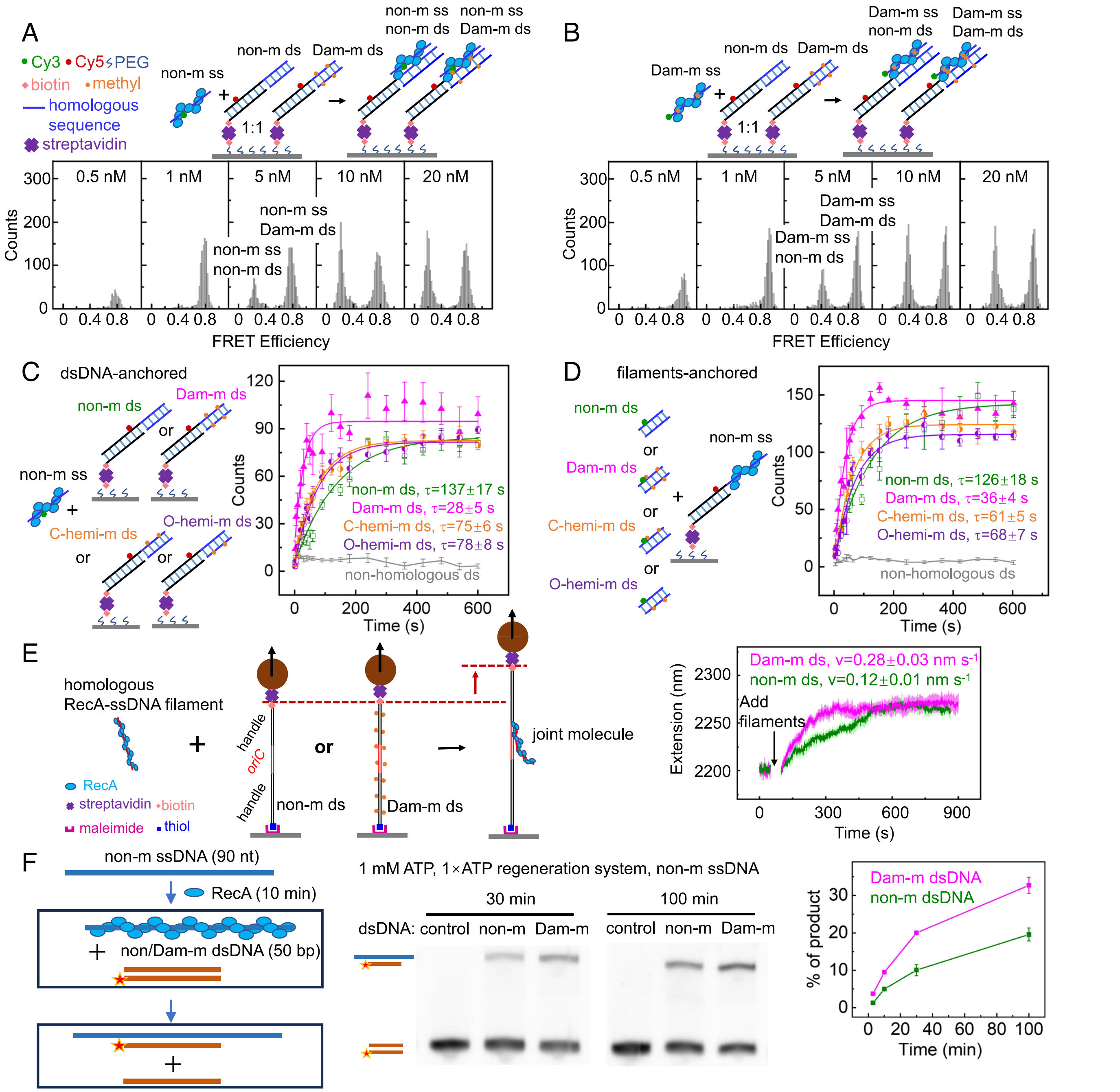

Fig. 2. Dam methylation of dsDNA accelerates homologous pairing, joint molecule growth, and strand exchange. (*A* and *B*) Equivalent glass surface-anchored non-m and Dam-m dsDNA compete for RecA-ssDNA filaments in solution, with non-m ssDNA used in (*A*) and Dam-m ssDNA used in (*B*). (*C* and *D*) The kinetics of homologous pairing, where dsDNA (*C*) or RecA-ssDNA filaments (*D*) were anchored to the glass surface. In (*A*–*C*), each anchored DNA contains a 21-bp homologous region, a 22-bp handle, and a 3-nt gap, while each ssDNA coated by RecA in solution is 21-nt. In (*D*), each anchored DNA contains a 21-nt homologous region, a 3-nt gap, and a 22-bp handle, while each dsDNA in solution is 21-bp. (*E*) Joint molecule growth studied by MT experiments. The dsDNA is anchored with a sequence containing a 412-bp homologous region with two handles, while the ssDNA coated by RecA in solution is 412-nt. (*F*) Bulk strand exchange assays with ATP. Control means the system without RecA. We used 1 mM ATPγS in (*A*–*E*). The error bars in panels (*C*–*F*) represent SE from more than three experiments.

opportunities for HR during repair. To address this concern, we assessed the relative level of DNA damage by analyzing plasmid topological forms using agarose gel electrophoresis. DNA damage leads to the relaxation of supercoiled plasmids, resulting in a shift toward open circular and linear forms. Therefore, a decrease in the supercoiled form indicates more DNA damage. As shown in *SI Appendix*, Fig. S15, plasmids extracted from WT (*EcoGII*) exhibited a more compact supercoiling distribution compared to WT, suggesting a lower rather than higher level of DNA damage than WT. Plasmids from the Δ*dam* strain showed a shift toward open circular and linear forms, indicating more DNA damage. These results demonstrate that the enhanced or impaired HR efficiency observed in WT (*EcoGII*) or Δ*dam* is not caused by increased or decreased DNA damage, but rather reflects a methylation-driven influence on HR.

As shown in Fig. 4, we next studied how DNA methylation affects cell sensitivity to ultraviolet radiation (UV), mitomycin C (MMC), methyl methanesulfonate (MMS), and phleomycin (PHL), which cause DNA damage and other effects. These agents have been commonly used to induce DSBs (65–67). As shown in the second row of Fig. 4*A*, Δ*dam* strain was hypersensitive to all four agents, consistent with previous studies (7, 9–11). As MMR

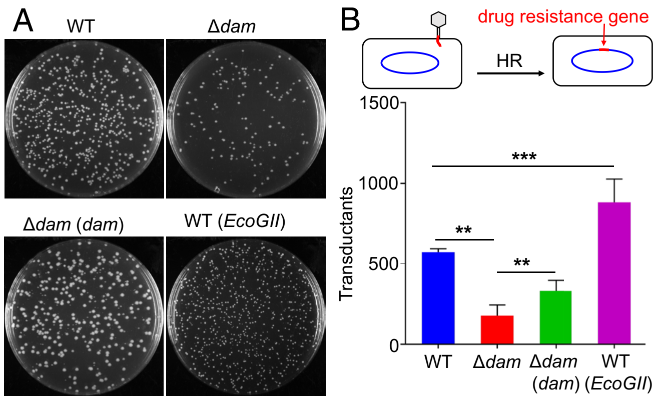

Fig. 3. P1 transduction assay to test the effects of DNA methylation on HR efficiency in cells. (*A*) Representative transduction results for the four strains indicated in the figure. (*B*) Averaged and SE of transductions from three replicates. The principle of the P1 transduction assay is sketched.

relies on Dam methylation, we deleted *mutH* from WT and Δ*dam* strains (referred to as Δ*mutH* and Δ*dam*Δ*mutH*) to abolish MMR and compared their sensitivities. We found both Δ*mutH* and Δ*dam*Δ*mutH* showed similar sensitivity to their MMR-reserving counterparts (WT and Δ*dam*), suggesting that MMR loss alone does not obviously increase sensitivity.

Combining the results that Δ*dam* has lower HR efficiency in P1 transduction, *dam* deletion did not significantly affect the expression of genes related to HR as shown the RNA-seq (*SI Appendix*, Table S2), we hypothesize that the preference of

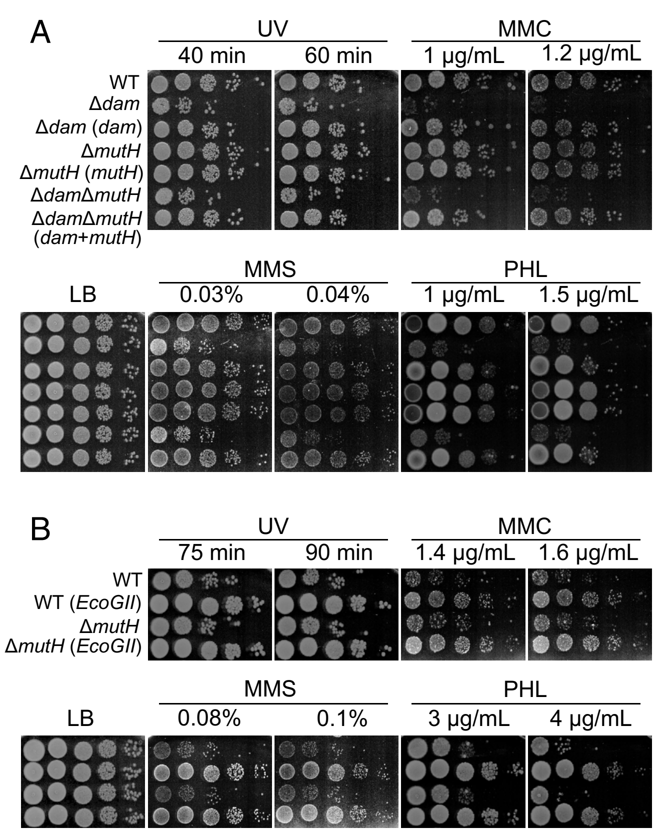

Fig. 4. Spot assays for UV, MMC, MMS, and PHL sensitivities of WT *Ec* and mutants. Cells were spotted in ten-fold serial dilutions and incubated at 37 °C for 24 h (MMC) or 16 h (others). Each condition was tested on three plates, with consistent results; one representative plate is shown per condition. (*A*) Spot assays for WT, *dam* deletion, *mutH* deletion, and double mutants. The deletions were rescued by introducing plasmid-borne genes. (*B*) Hypermethylated strains WT (*EcoGII*) and Δ*mutH* (*EcoGII*) exhibited even greater drug resistance than WT and Δ*mutH* strains, respectively under heavy doses of agents.

RecA for Dam-methylated DNA in RecA assembly, homologous pairing, growth of the joint molecule and strand exchange may contribute to the efficiency of HR. Previous studies have shown that double-deletion of *dam* and an HR-related gene causes difficulty for the *Ec* cells to survive (7, 10–13), which implies that the deletion of *dam* leads to DNA damage requiring HR to repair. Our hypothesis is not in conflict with these previous results.

We found the hypermethylated WT (*EcoGII*) and Δ*mutH* (*EcoGII*) showed even stronger drug resistance (about 100-fold surviving cells) than WT and Δ*mutH* strains, respectively (Fig. 4*B*). Notably, the hypermethylated strain WT (*EcoGII*) demonstrated the highest HR efficiency in the P1 transduction assay as well. These results suggest DNA methylation enhances RecA-mediated HR repair in cells, even though hypermethylation of the genome significantly decreases the growth rate (*SI Appendix*, Fig. S16). In RNA-seq, we also found that the overexpression of *EcoGII* does not significantly affect the expression of HR-related genes (*SI Appendix*, Table S2), which supports the hypothesis that the preference of RecA for methylated DNA may enhance HR.

Since Dam methylation affects the expression of many genes (68), we analyzed the regulation of genes using RNA-seq. We found *recA* was slightly upregulated (about 1.4-fold) rather than downregulated in Δ*dam*, which agrees with previous findings (7, 8, 12). After the *dam* gene was knocked out, significantly differentially expressed genes are predominantly leading to severely reduced acid stress resistance, significantly enhanced tryptophan and carbon metabolism pathways, and widespread metabolic remodeling. After overexpression of *EcoGII*, the significantly differentially expressed genes mainly disrupt genome-wide methylation patterns, driving cells into a “survival-first” state dominated by stress defense but accompanied by the risk of reduced genomic stability (*SI Appendix*, *Supplementary Discussion* and Dataset S1). We do not exclude the indirect effects of the above gene regulations on HR. Notably, the overexpression of *EcoGII* led to the downregulation genes related to energy metabolism and membrane transport as well as membrane motility, further contributing to a state of metabolic depression and slow growth of strains. These molecular-level alterations are consistent with our experimentally observed slow-growth phenotype of WT (*EcoGII*) (*SI Appendix*, Fig. S16).

Conserved Effects on RecA and Impacts on Molecular Motors. To test whether the preference of RecA for methylated DNA is conserved in RecA assembly and homologous pairing, we repeated single-molecule experiments with RecA from another gram-negative bacterium *Klebsiella pneumoniae* (KpRecA), a gram-positive bacterium *Bacillus subtilis* (BsRecA), and a flowering plant *Arabidopsis thaliana* (AtRecA2). In MT experiments, KpRecA, BsRecA, and AtRecA2 assembled faster on Dam-m ssDNA and even faster on m6A-m ssDNA (Fig. 5 *A*, *C*, and *E*). Monte Carlo calculations quantified how methylation accelerated nucleation and growth (*SI Appendix*, Table S1). The smFRET experiments showed that RecA-ssDNA filaments paired faster with Dam-m dsDNA (Fig. 5 *B*, *D*, and *E*). These results suggest the effects of DNA methylation on RecA assembly and homologous pairing are conserved across species.

The RecA-like domains are found in many molecular motors such as helicases, SMC proteins, and translocases, which use ATPase activity to convert chemical energy into mechanical motion along DNA (69). It would be interesting to explore whether Dam methylation affects the efficiency of these motors. We found RecQ, an SF2 family helicase involved in recombination and DNA repair, showed the ATPase activity increased by about

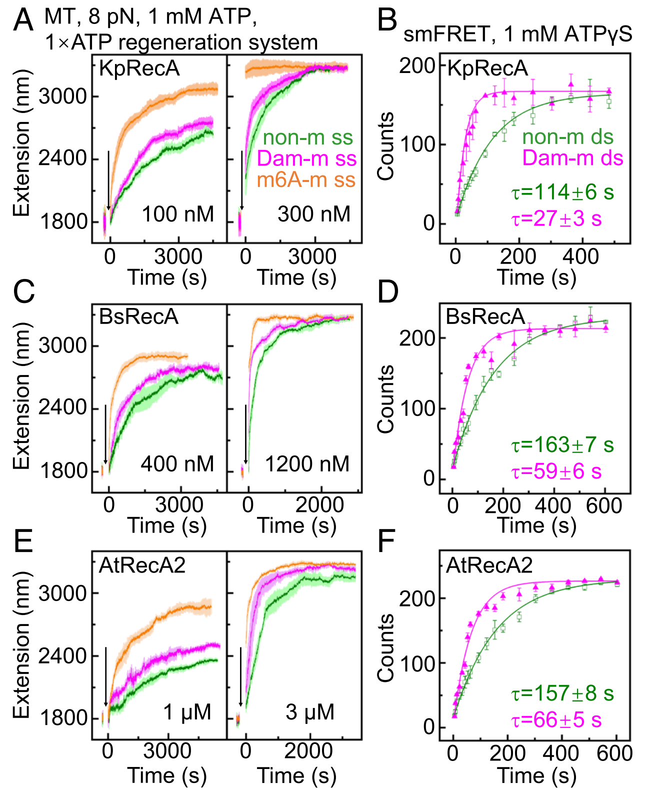

Fig. 5. Conserved preference of RecA for Dam-m DNA. The assembly of RecA on 6 k-nt ssDNA and the kinetics of homologous pairing mediated by RecA are shown for KpRecA (*A* and *B*), BsRecA (*C* and *D*), and AtRecA2 (*E* and *F*), respectively. The black arrow indicates the addition of RecA for assembly. Filaments were added to the flow cell where either non-m or Dam-m dsDNA was anchored to the surface for homologous pairing. The error bars represent SE from more than three experiments.

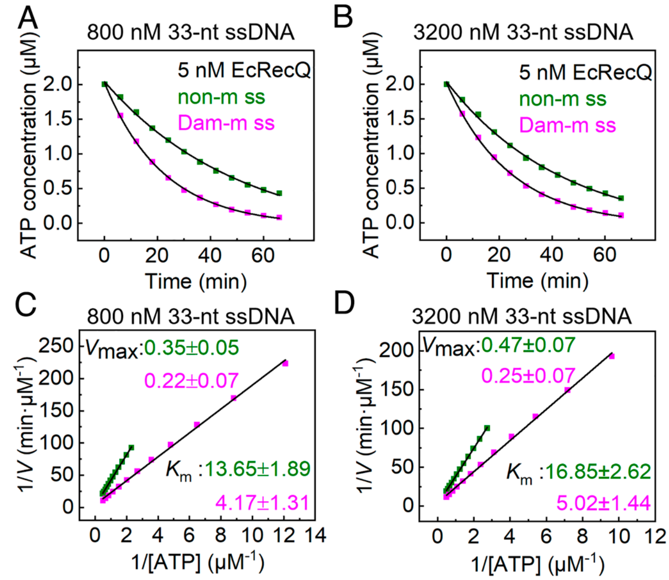

Fig. 6. Dam methylation of ssDNA promotes the ATPase activity of EcRecQ. (*A* and *B*) EcRecQ hydrolyzed ATP to translocate on ssDNA, resulting in a decrease in the concentration of the ATP. (*C* and *D*) We used the Lineweaver–Burk plot to determine the apparent affinity: K_m (μM) and the maximal velocity of ATP hydrolysis: V_{max} (μM·min^{-1}) with non-m ssDNA (olive) and Dam-m ssDNA (magenta) used in the experiment.

3.3-fold for 33-nt Dam-m ssDNA containing three GATC sites (Fig. 6).

Discussion

Our in vitro experiments show that Dam methylation of ssDNA promotes RecA-ssDNA filaments assembly both in nucleation and growth stages (Fig. 1). Dam methylation of dsDNA promotes homologous pairing, joint molecule growth, as well as strand exchange (Fig. 2). The preference of RecA for methylated DNA is conserved across divergent species (Fig. 5). In cells, P1 transduction assay shows that Dam methylation promotes HR (Fig. 3), which is consistent with the cell sensitivity tests using DNA-damaging agents (Fig. 4). Notably, hypermethylating the genome further enhances HR in P1 transduction and increases resistance to DNA-damaging agents. Dam methylation of ssDNA increases the ATPase activity of RecQ helicase that containing RecA-like domains (Fig. 6). Future studies could investigate whether this effect is common in motors with RecA-like domains.

To determine whether the enhancement of m6A on RecA assembly and homology pairing is specific to methylation at Dam sites, we designed ssDNA and dsDNA substrates in which GATC sites were mutated, thereby eliminating the canonical Dam recognition sequence. Notably, methylation of a single adenine the modified 20-nt sequence still promoted RecA nucleation in BLI experiments (*SI Appendix*, Fig. S17*A*), as evidenced by an about 1.4-fold increase in k_{on} (compared to about 2.0-fold for Dam methylation). In smFRET experiments (*SI Appendix*, Fig. S17*B*), after methylating the adenines in the modified sequence where the number of methylation sites is controlled, we observed an about 3.0-fold increase in RecA-mediated homologous pairing velocity (compared to about 3.5-fold for Dam methylation). Although these effects remain significant on RecA assembly and homologous pairing for the mutated sequences, they are less pronounced than those observed with Dam methylation, suggesting that the sequence context may influence the stimulatory effect of m6A but is not strictly required for promoting RecA activity.

One concern is that GATC sites occur only roughly every 200 base pairs throughout the genome, potentially limiting the biological relevance of Dam methylation. However, our MT experiments demonstrate that even with relatively low-density GATC sites (22 GATC sites in a 6 k-nt ssDNA), Dam methylation consistently accelerates RecA assembly on SSB-coated ssDNA by about 2.0-fold under all tested conditions (ATP, ATPγS, and ATP plus RecOR). Monte Carlo calculations further confirm that Dam methylation promotes both nucleation and growth of RecA filaments by approximately 2.0-fold. In BLI experiments, methylation at a single GATC site on the 20-nt ssDNA increased k_{on} by about 1.8-fold.

GATC sites are not uniformly distributed; notably, the *oriC* region has a GATC site density more than ten times higher than in random genomic regions. About 85% of the Chi sites, known HR hotspots, are concentrated within the 136-kb genomic region surrounding *oriC* (70). RecA preferentially assembles on DNA regions enriched in Chi sites (51, 71), and the overlap between Chi sites and densely distributed GATC sites around *oriC* suggests a possible link between Dam methylation and HR. Our data show a positive but nonlinear relationship between RecA assembly and the density of methylated adenines on ssDNA. In MT experiments using naked ssDNA (Fig. 1*E*), Dam methylation increased nucleation and growth rates by 2.0-fold and 1.9-fold, respectively, whereas m6A methylation led to greater increases of 5.1-fold and 3.2-fold. Similar results were consistently observed under other conditions, including SSB-coated ssDNA in the presence of ATP (Fig. 1*B*), ATPγS (Fig. 1*C*), and ATP plus RecOR (Fig. 1*D*), with m6A methylation consistently promoting faster assembly. Further MT experiments using a 2.5 k-nt ssDNA with higher GATC density (108 GATC sites) demonstrated a greater enhancement in nucleation (3.0-fold) and growth (2.7-fold), confirming the relationship between methylation density and RecA assembly efficiency (*SI Appendix*, Fig. S18*A*). In BLI experiments, methylation of a single GATC site in 20-nt ssDNA increased RecA nucleation rate (k_{on}) by about 1.8-fold, while methylation of all five adenines in the same sequence resulted in a more substantial increase (about 3.9-fold) (Fig. 1*F* and *SI Appendix,* Fig. S6). Likewise, Dam methylation of a 33-nt ssDNA containing three GATC sites resulted in an approximately 4.1-fold increase in k_{on} (*SI Appendix*, Fig. S18*B*).

Previous studies have shown that the biophysical properties of DNA influence RecA assembly. RecA prefers poly (dT) over poly (dA) and poly (dC) because of its high flexibility (72) and binds more tightly to pyrimidine-rich sequences (51, 73), while purine-rich sequences hinder assembly due to increased base stacking. Previous works also reported that modifications and drug intercalation affected the assembly of RecA on dsDNA (74–78). The negative charge of ssDNA hinders RecA assembly through electrostatic repulsion with the negatively charged C-terminus of RecA (57, 58, 72, 79). In this work, we find that Dam methylation accelerates RecA assembly (Fig. 1) partially by reducing the effective charge of ssDNA under counterion screening and thereby mitigating electrostatic repulsion between RecA and ssDNA (Fig. 1 *G*–*I*).

In addition to Dam-mediated adenine methylation at GATC sites, the most frequent cytosine methylation in *Ec* is catalyzed by the Dcm (DNA cytosine methyltransferase) enzyme, which targets the internal cytosine of CCWGG motifs (where W = A or T), generating 5-methylcytosine (m5C). Transcriptomic analyses have demonstrated that cytosine methylation modulates gene expression in *Ec* (80–82). Since adenine methylation enhances RecA assembly, we further examined whether cytosine methylation can also affect RecA assembly. In MT experiments, we compared RecA assembly kinetics on naked 6 k-nt ssDNA (which has the same sequence as that used in Figs. 1 and 5) in which all the 1,582 cytosines were either methylated (m5C-m) or unmethylated (non-m). Under conditions of 200 nM RecA, 1 mM ATP, m5C methylation had no significant effect on RecA filament formation, as indicated by similar nucleation and growth rates between m5C-m and non-m ssDNA (*SI Appendix*, Fig. S19*A*). These results are consistent with electrophoretic mobility assays, which showed that cytosine methylation does not obviously alter the effective charge of ssDNA under counterion screening (*SI Appendix*, Fig. S19*B*).

In all, our findings suggest that Dam methylation may enhance HR by promoting RecA assembly, homologous pairing, joint molecule growth, and strand exchange. Future theoretical and computational studies could clarify how Dam methylation influences the functions of RecA, and fluorescence experiments may reveal real-time effects of Dam methylation on HR in vivo. Overall, our study provides insights into how cells maintain genomic integrity and unique functions of Dam methylation.

Materials and Methods

Protein Purification. We expressed and purified the proteins using a His-SUMO-tag system in *Ec*, followed by Ni-NTA affinity, SUMO tag cleavage, ion exchange, and gel filtration chromatography (*SI Appendix*, Fig. S20).

Reaction Conditions. All in vitro assays were performed in RecA reaction buffer (25 mM Tris-Ac, 100 mM NaAc, 10 mM $MgAc_2$, pH 7.5). When 1 mM ATP was used, we used the ATP regeneration system contained 20 mM phosphoenolpyruvate, 100 U/mL pyruvate kinase, and 10 mM KCl. In smFRET experiments, an oxygen scavenging system (0.8% glucose, 1 mg/mL glucose oxidase, 0.4 mg/mL catalase, and 1 mM Trolox) was added to the buffer. In BLI experiments, 0.02% BSA, 0.05% Tween-20, 1 mM DTT were added to the buffer. MT and smFRET assays were conducted at 22 °C, BLI at 30 °C, and bulk strand exchange and ATPase assays at 37 °C. We listed the sequences of oligos used in smFRET, BLI, native PAGE, and ATPase activity experiments (*SI Appendix*, Table S3).

MT Experiments. The MT setup was built on an inverted microscope with a 40 × oil immersion objective. Flow cells were assembled from APTES-functionalized glass slides and modified with sulfo-SMCC to immobilize thiol-labeled DNA. Streptavidin-coated superparamagnetic beads (Dynabeads M-270) were bound to the biotinylated DNA ends. A pair of NdFeB magnets, driven by a stepper motor, was used to apply constant stretching forces to the beads, and bead positions were monitored in three dimensions using CCD imaging. The DNA extension was determined as the distance between the bead and the glass slide. Please see *SI Appendix*, Figs. S21–S23 details to make DNA samples for MT experiments.

BLI Experiments. BLI experiments were performed using the Octet RED96e system. Streptavidin biosensors were pre-equilibrated in buffer. Biotinylated short ssDNA oligos were immobilized on the streptavidin biosensors, followed by association with RecA and subsequent dissociation in buffer. Kinetic constants were obtained using the Octet Data Analysis software.

smFRET Experiments. The smFRET experiments were conducted on a custom-built TIRF microscope equipped with 532 and 640 nm lasers and an EMCCD camera. Biotinylated DNA substrates were immobilized onto PEG-biotin-coated slides via streptavidin.

Strand Exchange Assays. For bulk strand exchange assays, RecA-ssDNA filaments were assembled on 90-nt ssDNA by incubating with RecA for 10 min. Strand exchange was initiated by adding Cy3-labeled 50-bp dsDNA and incubating for 30 min or for other durations as indicated, then terminated by SDS/proteinase K treatment. Reaction products were resolved on 10% native PAGE, imaged using a Typhoon scanner, and quantified with ImageJ.

ATPase Activity Assay. The ATP hydrolysis by EcRecQ was measured using a luminescent kinase assay kit (Beyotime) according to the manufacturer's protocol. ATP consumption was monitored over time using a microplate reader, and kinetic parameters (K_m and V_{max}) were determined using Lineweaver–Burk plots.

Cellular Experiments and RNA-seq. *Ec* strains were cultured in LB medium. P1 transduction assays were conducted using standard protocols with zapB::kan donor phage, and transductions were selected on antibiotic-containing plates with arabinose induction where needed. DNA damage sensitivity was tested by spotting serial dilutions of cultures onto LB agar with or without genotoxic agents (e.g., UV, MMC) and incubating for 16 to 24 h. For RNA-seq, WT and Δ*dam* or *EcoGII*-overexpressing cells were treated with 8 μg/mL MMC for 40 min during log-phase growth. Total RNA was extracted and sequenced on the Illumina NovaSeq 6000 platform. Differential gene expression analysis was performed using DESeq2. Please see *SI Appendix,* Tables S4–S6 for details on *Ec* strains, plasmids, and oligos in cellular experiments.

Please see *SI Appendix*, *Materials and Methods* for more details.

Data, Materials, and Software Availability. All study data are included in the article, supporting information and/or supporting datasheet.

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